K.D.S.H.M. Perera

B.Sc. Engineering (Honours)

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Profile

My research lies at the intersection of bio-inspired mechanisms and soft robotics, with a focus on the dynamic modeling and control of unconventional locomotion systems. I investigate how biological principles can inform the morphology, control, and sensing of adaptive robotic mechanisms, and how these systems can be scaled into swarm frameworks for robust, decentralized operation in unstructured environments.

Education

2021 – 2025 **B.Sc. Engineering (Hons) in Mechanical Engineering**
Specialisation: Mechatronic Systems Engineering, *University of Moratuwa*, Sri Lanka
GPA: 3.79 / 4.00
Class Rank: 10/120
Minor: Mathematics

2017 – 2019 **GCE Advanced Level, Physical Science Stream**, *D.S. Senanayake College, Colombo 07*
Results: 2 A's & 1 B; Z-Score: 2.3409

2011 – 2016 **GCE Ordinary Level**, *D.S. Senanayake College, Colombo 07*
Results: 8 A's & 1 B

Work Experience

Feb 2025 – Present **Temporary Lecturer**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Aug 2025 – Jan 2026 **Temporary Instructor**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Mar 2024 – May 2024 **Intern / Student Researcher**, *Shandong University of Science and Technology*, Shandong, China

Nov 2023 – Feb 2024 **Intern / Project Engineer**, *Dipped Product PLC*, Sri Lanka

Academic Projects

Underwater Drone for Coral Observation — Contributed to the design and development of a shallow-sea underwater drone using a novel fuzzy-adaptive sliding-mode controller for robust trajectory tracking; developed a VR-based teleoperation interface for real-time control and visualization.

Brachiating Robot — Designed, simulated (Gazebo), and built a prototype brachiation robot to investigate dynamic locomotion and control across variable bar spacing and heights.

Environment Monitoring Station — Implemented real-time environmental monitoring with signal conditioning, processing, remote communication and data visualization (web/mobile).

App-Controlled Mobile Robot	Developed a differential-drive mobile robot with image-based object detection, path planning, and manual/autonomous operation via a mobile app.
Micro-ROS Based Home Automation	Led development of an IoT smart-home prototype integrating micro-ROS for autonomous temperature-based fan control and web-based lighting/fan management.
Smart Garbage Collector	Collaboratively engineered a mobile robot using image processing to autonomously navigate predefined routes and collect litter.
Modular Microconveyor	Designed a modular microconveyor using chevron electrothermal actuators and a serpentine beam ratchet for precise, incremental motion; validated with multiphysics simulation.

Industrial Projects

Dynamic Work Bar Transfer Mechanism	Designed a pneumatic system to halt and transfer work bars between leaching tanks.
Web-Based SCADA Visualization	Developed a full-stack web application for real-time SCADA dashboards and historical data visualization.
Hydraulic Lifting Platform	Researched and prototyped a hydraulic lift for wheelchair accessibility (Shandong University of Science and Technology).

Achievements

Best Paper Award	Robotics & Intelligent Systems Track — Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka.
The Institution Best Project	Institution of Mechanical Engineers (IMechE) — University of Moratuwa, Sri Lanka (2025).
Dean's List	Semesters 1, 2, 5, 7

Research Publications

- (First author & presenter) *Brachiation Robot Inspired by Gibbon Dynamics: Insight for Control Across Irregular Bar Spacing and Heights*. Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka. **Best Paper Award, Robotics & Intelligent Systems**.DOI: 10.1109/MERCon67903.2025.11217001.
- (First author & presenter) *A Fuzzy-Adaptive Sliding-Mode Control Strategy for Optimizing Control of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka.DOI: 10.1109/MERCon67903.2025.11217001.
- (Co-first author) *Development of Marine Grade Acrylonitrile – Butadiene – Styrene (ABS) based Hull of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka. DOI: 10.1109/MERCon67903.2025.11217039.
- (Co-first author) *CFD-Driven Hydrodynamic Optimisation of a Whale Shark–Inspired Underwater Drone for Shallow Reef Observation*. 12th International Conference on Heat Transfer and Fluid Flow (HTFF 2025). DOI:10.11159/htff25.243.
- (First author) *A Chevron-Actuated Microconveyor for Precise Wafer-Scale Microfactory Transportation*. Mechanical Engineering Research Symposium 2025 (MERS 2025), Department of Mechanical Engineering, University of Moratuwa. Accepted.
- (Co-first author) *WS Ray: Extending Human Perception with a Teleoperated Underwater Drone*

for Subaquatic Environments.International Conference on Automation Science and Engineering (CASE 2026). Under review.

Teaching & Mentorship Experience

Temporary Instructor/Lecturer	Designed and delivered lectures on **Robot Operating System (ROS 2)** and **modeling & simulation of mechatronic systems**.
	Developed and conducted hands-on ROS 2 workshops for undergraduate students
	Mentor two university teams developing an Autonomous Underwater Vehicle (AUV) for the Singapore AUV Challenge (SAUVC 2026).
	Designed, developed, and conducted laboratory modules for Mechanical & Mechatronic Engineering courses

Skills

Programming	Python, C++, MATLAB & Simulink , LabVIEW
Robotics & Simulation	ROS2, micro-ROS, Gazebo, MuJoCo, Webots, OpenCV
Design & Analysis	SolidWorks, AutoCAD, Ansys, Adams, Fluidsim
Embedded Systems & Electronics	Atmel Studio, Proteus, Arduino & ESP32, Raspberry Pi
Web & Data	HTML/CSS, JavaScript, SQL, Flutter
Office & Graphics	LaTeX, Microsoft Word, PowerPoint, Excel, Adobe Illustrator, Photoshop

Languages

Sinhala	Native
English	Fluent

Co-curricular Activities

Committee Member	EXMO 2023 Logistics, University of Moratuwa
Committee Member	MechatronX 2023, Inter-School Robotics and Automation Competition
Society Memberships	IMechE, UMMES
School Activities	Science Quiz Team Organizer, D.S. Senanayake College
Community Service	Danuma Yathra Member (2019–2022)

References

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University of Moratuwa
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